

PS

2

CSCI 4220 • MACHINE & DEEP LEARNING

The Cat Detector

Week 3–5 • Eight problems, one cat, one very long week

Due Sun 11 Oct, 11:59pm • 100 points • Submit a single PDF to Canvas

It's just a flesh wound.

— The Black Knight, *Monty Python and the Holy Grail*, 1975

HOW THIS WORKS

On Monday morning Robby's cat walked across his keyboard during a faculty Zoom meeting. By Sunday night Robby had built a deep network to catch it, trained it, broken it, fixed it, and been quizzed about it at brunch. One problem per day, and two on Sunday.

Every problem starts with **one worked example** of the same kind of step, done in full with different numbers. Copy the shape. No code and no calculator: the few sigmoids and logarithms you need are below.

THE WEEK

Mon	The cat joins the faculty meeting	a deep network, forward	15
Tue	The index cards	the training loop	12
Wed	Swagger and Hedge	confidence and cross-entropy	12
Thu	Dr. Dan's sticky note	accuracy, precision, recall	15
Fri	Four runs, no labels	reading loss curves	14
Sat	The agent that aced the test	validation vs. test	12
Sun	Brunch with Dr. Sanders	multiple choice	12
Sun	Explain it to the cat	free response	8

NUMBERS YOU MAY NEED

z	-2	-1	0	1	3	p	0.01	0.1	0.4	0.9
$\sigma(z)$	0.12	0.27	0.50	0.73	0.95	$\ln p$	-4.61	-2.30	-0.92	-0.11

Submit one PDF to Canvas. Typed or neatly handwritten and scanned are both fine — legibility is graded, handwriting is not. You may discuss the problems with classmates, but write your solutions alone. AI tools are permitted and encouraged; if you use one, say so briefly at the top and say what it did for you.

1 Monday 9:02 a.m. — The cat joins the faculty meeting

BY HAND 15 pts

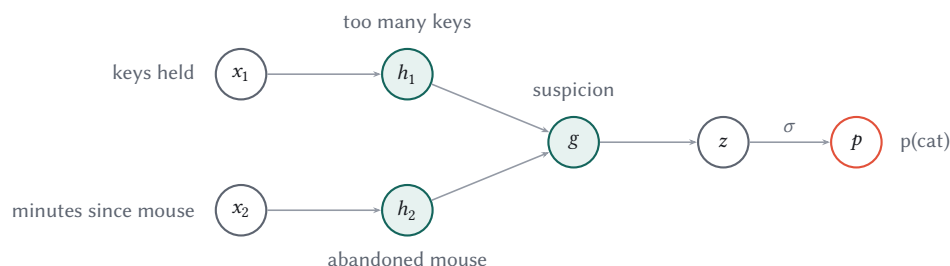
Nine minutes into the faculty meeting, Robby's cat walks the length of his keyboard. It unmutes him, types ffffffffffffffff; into the chat, and raises his hand. Dr. Dan calls on him. Robby has no idea what the question was, and decides the cat needs a detector.

The cat does two things Robby never does: it holds down lots of keys at once (x_1 = keys held down), and it never touches the mouse (x_2 = minutes since the mouse last moved). Robby's network has two hidden layers of ReLU units, $a[z] = \max(0, z)$, each with a job:

layer 1: $h_1 = a[x_1 - 2]$ ("too many keys") $h_2 = a[x_2 - 1]$ ("abandoned mouse")

layer 2: $g = a[h_1 + h_2 - 1]$ ("suspicion")

output: $z = g - 2$, $p(\text{cat}) = \sigma(z)$



Two inputs, two ReLU units in layer 1, one in layer 2, and a sigmoid that turns the final number into a probability.

✎ HERE'S ONE DONE FOR YOU

Robby doing a four-key shortcut he found online: $x = (4, 1)$.

$$h_1 = a[4-2] = 2, \quad h_2 = a[1-1] = 0, \quad g = a[2+0-1] = 1, \quad z = 1-2 = -1, \quad p(\text{cat}) = \sigma(-1) \approx 0.27.$$

Probably not the cat. One layer at a time: add things up, then ReLU turns anything negative into 0.

- The cat, standing on six keys, mouse untouched for three minutes: $x = (6, 3)$. Find h_1 , h_2 , g , z and $p(\text{cat})$.
- Robby typing normally: $x = (1, 0)$. Find the same five numbers. Which hidden units are off (equal to 0)?
- The cat stands on six keys *and* bats the mouse: $x = (6, 0)$. Is it still caught?
- In IC 4 every unit was connected to every unit in the layer before. If this network were built that way, with layer sizes $[2, 2, 1, 1]$, how many parameters (weights plus biases) would it have?

PROBLEM 1 • YOUR WORK

2 Tuesday 7:30 a.m. — The index cards

SHORT ANSWER 12 pts

Robby wrote the four steps of his training loop on index cards and taped them above his desk. Overnight the cat knocked them all onto the floor. They now read, in the order the cat left them:

- A Update:** move every weight a small step against its gradient.
- B Predict:** run the network forward on the training data.
- C Blame:** backpropagate to get the gradient of the loss for every weight.
- D Measure:** compute the loss.

The update rule is the one from PS1: $w \leftarrow w - \eta \cdot \text{gradient}$.

 HERE'S ONE DONE FOR YOU

One weight, one step. A weight $w = 1.0$ has gradient 3, and the learning rate is $\eta = 0.1$:

$$w \leftarrow 1.0 - 0.1 \times 3 = 0.7.$$

A *positive* gradient means making w bigger would make the loss bigger, so the step makes w smaller.

- (a) Put the cards in the right order. What happens after the last card?
- (b) One update with $\eta = 0.1$. Find the three new weights:
- | | | | |
|------------|-----|------|-----|
| weight now | 0.5 | -1.0 | 2.0 |
| gradient | 2 | -4 | 0 |
- (c) The third weight's gradient is 0 on every step. What happens to that weight over training? In a ReLU network, what could cause this?

BRING OUT YOUR DEAD

A ReLU unit that outputs 0 for every input passes back no gradient, so its weights stop changing — forever. It is called a *dead* unit. In IC 4, one oversized learning rate killed every unit at once.

PROBLEM 2 • YOUR WORK

3 Wednesday 3:40 p.m. — Swagger and Hedge

BY HAND 12 pts

Robby now has two detectors and cannot choose. **Swagger** is never less than 99% sure of anything. **Hedge** has never been more than 60% sure of anything. Each outputs $p = p(\text{cat})$. Four test clips:

clip	1	2	3	4
truth	cat	no cat	no cat	cat
Swagger's p	0.99	0.99	0.01	0.99
Hedge's p	0.6	0.6	0.4	0.6

Networks like these are trained with the **cross-entropy** loss (UDL ch. 5). For one clip: if the cat *was* there the loss is $-\ln p$; if it was *not*, the loss is $-\ln(1 - p)$.

 HERE'S ONE DONE FOR YOU

A detector says $p = 0.9$. If the cat was there: loss = $-\ln 0.9 \approx 0.11$ – right and confident, cheap. If the cat was not there: loss = $-\ln(1 - 0.9) = -\ln 0.1 \approx 2.30$ – wrong and confident, about twenty times as expensive.

- Say “cat” whenever $p > 0.5$. What is each detector’s accuracy on the four clips?
- Both detectors get the same clip wrong. Which clip, and what is each detector’s loss on it?
- Which detector would you trust, and what does the loss notice that accuracy does not?

IT'S ONLY A MODEL

One of our research talks asked whether large language models know what they don't know. Same idea: a model that is sure and wrong is worse than one that is unsure and wrong, because people act on its confidence.

PROBLEM 3 • YOUR WORK

4 Thursday 10:30 a.m. — Dr. Dan's sticky note

BY HAND 15 pts

Robby runs his detector on 100 one-minute clips of his desk. The cat was on the keyboard in 10 of them.

	cat really there	no cat
detector says cat	8 (true positives)	4 (false positives)
detector says no cat	2 (false negatives)	86 (true negatives)

Robby announces his accuracy at the faculty meeting. Dr. Dan holds up a sticky note that says NO CAT, points out that it is almost as accurate, and mentions that it took him forty seconds to build.

 HERE'S ONE DONE FOR YOU

A spam filter on 20 emails: 3 true positives, 1 false negative, 2 false positives, 14 true negatives.

$$\text{accuracy} = \frac{3+14}{20} = 85\%, \quad \text{precision} = \frac{TP}{TP+FP} = \frac{3}{5}, \quad \text{recall} = \frac{TP}{TP+FN} = \frac{3}{4}.$$

Precision: of the emails it flagged, how many were really spam. *Recall*: of the real spam, how much it caught.

- What is the accuracy of Dr. Dan's sticky note? How many of the 10 cats does it catch?
- Compute Robby's accuracy, precision and recall.
- A missed cat can email the dean. A false alarm just means Robby glances at his desk. Which number – precision or recall – should Robby try hardest to raise, and why?

YOUR MOTHER WAS A HAMSTER

When one class is rare, accuracy mostly measures how rare it is. Check what the do-nothing answer scores before you celebrate.

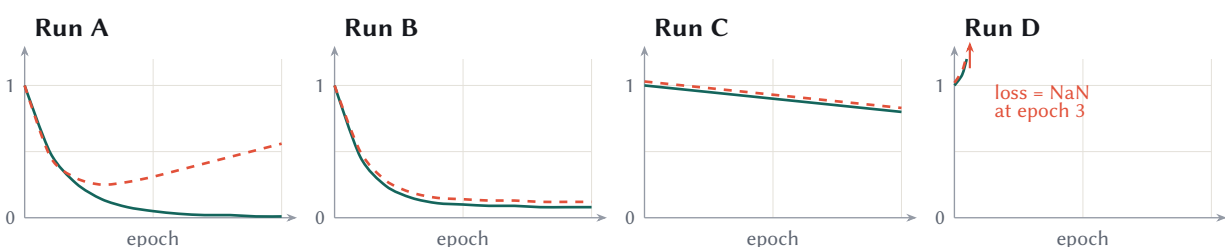
PROBLEM 4 • YOUR WORK

5 Friday 7:10 a.m. — Four runs, no labels

READ THE PLOTS 14 pts

On Thursday night Robby started four training runs and wrote their settings in a notebook. On Friday the cat is asleep on the notebook, and Robby is not going to be the one who moves her. He remembers the four settings, but not which run was which:

1. learning rate 1.0
2. learning rate 0.00001
3. learning rate 0.001, only 60 training clips, a huge network
4. learning rate 0.001, 1,000 training clips, a right-sized network



Solid: training loss. Dashed: validation loss (clips the run never trained on).

✎ HERE'S ONE DONE FOR YOU

A fifth curve, not in the figure: training and validation loss both fall a little and then flatten, side by side, at a high loss. They agree, so the network isn't memorising anything — it just can't do better. That is *underfitting*: the network is too small for the job.

- (a) Match runs A–D to settings 1–4, with a few words of evidence each.
- (b) One run's validation loss turns back up. Robby saved a copy of that network every 5 epochs; its validation loss was
- | | | | | | | |
|-----------------|------|------|------|------|------|------|
| epoch | 5 | 10 | 15 | 20 | 25 | 30 |
| validation loss | 0.46 | 0.30 | 0.25 | 0.27 | 0.31 | 0.36 |
- Which copy should he keep?
- (c) Name one change to that run's setup that would help, and say why.

PROBLEM 5 • YOUR WORK

6 Saturday 11:00 a.m. — The agent that aced the test

SHORT ANSWER 12 pts

After a research talk in class asking whether AI agents can do machine learning research, Robby lets one tune his detector while he goes swimming. When he gets back, the agent has left a note: “BEST TEST ACCURACY: 97.5%!!!”

The log says it tried every combination of learning rate $\in \{0.01, 0.001, 0.0001\}$ and width $\in \{32, 64, 128, 256\}$, trained each one twice (two random seeds), at 5 minutes per run. It scored every run on the **test** set and kept the best. Its four finalists:

network	validation accuracy	test accuracy
P	94.0%	95.5%
Q	96.5%	94.0%
R	95.0%	97.5%
S	93.0%	96.0%

HERE'S ONE DONE FOR YOU

The honest way. Split 1,000 clips into 700 for *training* (fit the weights), 150 for *validation* (choose the settings), and 150 for *testing*. Try every setting, pick the one with the best **validation** score, then open the test set **once**, for that one network, and report whatever it says.

- (a) How many combinations did the agent try, how many training runs was that, and how long did they take?
- (b) Which network should have been chosen, and what accuracy should have been reported?
- (c) In one or two sentences: why is “97.5%” too good to be true?

NOBODY EXPECTS THE TEST SET

The test set is a sealed envelope. Once you use it to make a choice, it stops being a fair measure of the thing you chose.

PROBLEM 6 · YOUR WORK

7 Sunday 10:00 a.m. — Brunch with Dr. Sanders

MULTIPLE CHOICE 12 pts

Dr. Sanders has heard about the cat detector and has prepared a quiz, which Robby must pass before anyone passes him the pancakes. Circle one answer for each. Two points each; no work needed.

 HERE'S ONE DONE FOR YOU

Which of these is a hyperparameter? (A) a weight in Ω_1 (B) the learning rate (C) a bias in β_0 (D) the output p .

Answer: (B). You choose the learning rate before training; the weights and biases are what training learns, and p is what the network computes.

- (a) What is $\text{ReLU}(-3)$?
 (A) 3 (B) -3 (C) 9 (D) 0
- (b) Training loss keeps falling, but validation loss starts rising. The network is...
 (A) underfitting (B) learning too slowly
 (C) overfitting (D) haunted
- (c) The loss becomes NaN after three steps. What should Robby try first?
 (A) a smaller learning rate (B) more training data
 (C) a deeper network (D) a new laptop
- (d) The detector outputs $p = 0.95$ for a clip. This means...
 (A) the detector is 95% accurate (B) its estimate that this clip has a cat is 0.95
 (C) 95% of the pixels are cat (D) the cat is 95% cat
- (e) In IC 4 a deep network beat a shallow one with the same number of parameters. Mostly because...
 (A) its layers fold and re-cut the picture into many more pieces (B) it had more parameters
 (C) it used a bigger learning rate (D) the cat preferred it
- (f) LoRA and DoRA (from one of our research talks) fine-tune a big network by...
 (A) retraining every weight from scratch (B) deleting layers
 (C) asking the cat (D) learning a small, low-rank change to frozen weights

PROBLEM 7 · YOUR WORK

8 Sunday 9:00 p.m. — Explain it to the cat

FREE RESPONSE 8 pts

The detector works. The cat, now unable to use the keyboard, sits on Robby's lap and stares at him. Robby decides to explain machine learning research to it.

Pick **one research paper presented in class so far**. In **150–250 words**, in plain language the cat could follow (no equations), explain:

1. the problem the paper tackles, and why it matters;
2. what the authors did that was new;
3. one limitation, or one way it could go wrong;
4. one way its idea could help (or hurt) Robby's cat detector.

 HERE'S ONE DONE FOR YOU, ON A PAPER WE DIDN'T PRESENT

Dropout (Srivastava et al., 2014). **Problem:** big networks memorise their training data and do worse on new data. **What's new:** during training, randomly switch off some units at every step, so no unit can rely on any other — like a team that practises with random players missing, so everyone has to be useful. **Limitation:** training takes longer, and it does nothing for a network that is too small to begin with. **For Robby:** Friday's Run A, the huge network memorising 60 clips, is exactly the problem dropout was built for.

PROBLEM 8 • YOUR WORK

